

Nepal PreTST 2025 Solutions

PRAKRIT GAJUREL

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§1 Problems

Problem 1.1. Shining tells Prajit a positive integer $n \geq 2025$. Prajit then tries to place n points such that no four points are concyclic and no 3 points are collinear in Euclidean plane, such that Shining cannot find a group of three points such that their circumcircle contains none of the other remaining points. Is he always able to do so?

Problem 1.2. (a) Positive rational numbers a, b , and c have the property that $\frac{a}{b} + \frac{b}{c} + \frac{c}{a}$ is an integer. Is it possible for $\frac{a}{c} + \frac{c}{b} + \frac{b}{a}$ to also be an integer except for the trivial solution?

(b) Positive real numbers a, b , and c have the property that $\frac{a}{b} + \frac{b}{c} + \frac{c}{a}$ is an integer. Is it possible for $\frac{a}{c} + \frac{c}{b} + \frac{b}{a}$ to also be an integer except for the trivial solution?

Problem 1.3. Let the incircle of $\triangle ABC$ touch sides BC , CA , and AB at points D , E , and F , respectively. Let D' be the diametrically opposite point of D with respect to the incircle. Let lines AD' and AD intersect the incircle again at X and Y , respectively. Prove that the lines DX , $D'Y$, and EF are concurrent, i.e., the lines intersect at the same point.

Problem 1.4. Find all pairs of positive integers n and x such that

$$1^n + 2^n + 3^n + \cdots + n^n = x!$$

§2 Solutions

§2.1 Problem 1, proposed by Shining Sun(USA) and Prajit Adhikari(Nepal)

Problem 2.1. Shining tells Prajit a positive integer $n \geq 2025$. Prajit then tries to place n points such that no four points are concyclic and no 3 points are collinear in Euclidean plane, such that Shining cannot find a group of three points such that their circumcircle contains none of the other remaining points. Is he always able to do so?

Proof.

□

§2.2 Problem 2, proposed by Andrew Brahms(USA)

Problem 2.2. (a) Positive rational numbers a, b , and c have the property that $\frac{a}{b} + \frac{b}{c} + \frac{c}{a}$ is an integer. Is it possible for $\frac{a}{c} + \frac{c}{b} + \frac{b}{a}$ to also be an integer except for the trivial solution?

(b) Positive real numbers a, b , and c have the property that $\frac{a}{b} + \frac{b}{c} + \frac{c}{a}$ is an integer. Is it possible for $\frac{a}{c} + \frac{c}{b} + \frac{b}{a}$ to also be an integer except for the trivial solution?

Proof. We solve part (b) first.

Remark 2.3. Note that the trivial solution is obviously $a = b = c$. However, it was mentioned in the contest that $a = b$ or any permutation of it is allowed, so we make full use of it below.

Claim 2.4 — For positive real numbers a, b, c , it is possible.

Proof. Set $a = b$ and we get that the first expression and the second expression are equal. Therefore, we need to find a value for a, b , and c such that the expression is equal. This is obviously quite easy, we can just set $a = b = 1$ and solve some equation by setting

$$1 + c + \frac{1}{c} = k$$

where k is some integer. For our example let $k = 4$, giving $c = \frac{3+\sqrt{5}}{2}$, which works upon checking and we're done. □

□

§2.3 Problem 3, proposed by Kritesh Dhakal(Nepal)

Problem 2.5. Let the incircle of $\triangle ABC$ touch sides BC , CA , and AB at points D , E , and F , respectively. Let D' be the diametrically opposite point of D with respect to the incircle. Let lines AD' and AD intersect the incircle again at X and Y , respectively. Prove that the lines DX , $D'Y$, and EF are concurrent, i.e., the lines intersect at the same point.

Proof. We define $T := D'Y \cap DX$, $Z := DD' \cap AT$, and denote by I the incenter of $\triangle ABC$. Further let ω denote the incircle.

Obviously DD' is a diameter to the ω , thus $D'Y \perp AD$ and $D'X \perp DT$, which also implies that $DZ \perp AT$. This implies that all the points A, F, I, E, Z lie on a circle with radius AI . Further, note $ADXZ$ is cyclic, so by the radical axis theorem on the circles $(ADXZ)$, $(AFIE)$, and ω we get that AZ , EF , DX concur at T . However as T is also the intersection of lines DX and $D'Y$, we get that EF , DX , and $D'Y$ concur at T , as desired. $\square$

§2.4 Problem 4, proposed by Petko Lazarov(Bulgaria)

Problem 2.6. Find all pairs of positive integers n and x such that

$$1^n + 2^n + 3^n + \cdots + n^n = x!$$

Proof.

□